

Supplementary Information

ReMol: A Chemical Reaction Knowledge-guided Self-supervised

Molecular Image Representation Learning Framework

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Table of Contents

Supplemental Materials and Methods	3
A. Supplementary Experiment Setup.....	3
A.1 Dataset details	3
A.2 Hyperparameters	5
B. Supplementary Methods	5
B.1 Time Complexity	5
B.2 Computational Requirements	7
C. Supplementary Results	8
C.1 Results on Pretraining.....	8
C.2 Statistical Significance Analysis	8
C.3 Dataset Bias Analysis	9
C.4 Few-shot Learning Analysis	10
C.5 Reaction-type Analysis	11
C.6 Comparison with Multimodal Methods	12
C.7 Ablation Study	13
C.8 Case Study	15
Supplementary References.....	20

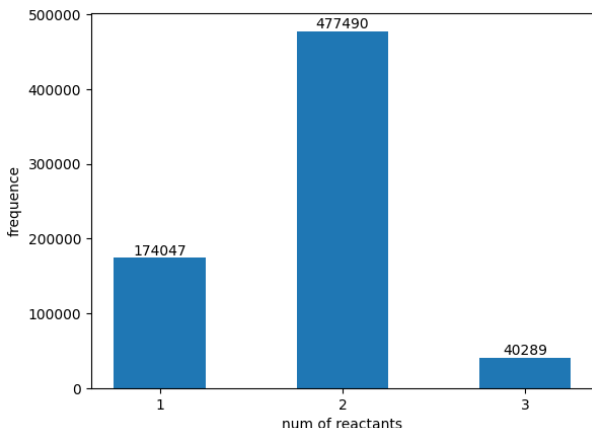
Supplemental Materials and Methods

A. Supplementary Experiment Setup

A.1 Dataset details

Pretraining Dataset

To pretraining ReMol, we first processed 1,013,118 reactions from USPTO-granted patents in the USPTO-full¹ dataset. We filtered the reactions that could not extract reaction templates and reactions that appeared in the downstream task test set to prevent data leakage. Finally, we obtained 691,826 reactions and divided them into a training set (80%) and a validation set (20%). As shown in the **Supplementary Figure 1**, we display the number of reactants in the pretraining dataset.



Supplementary Figure 1. Statistics of the number of reactants in the pretraining data.

Fine-tuning Dataset

As shown in the **Supplementary Table 1**, we display the details of the fine-tuning dataset. For the reaction product prediction task (**Table I**), chemical reaction classification (**Table II**), and molecular property prediction benchmarks (**TABLE III** and **TABLE IV**), we conducted experiments using three different random seeds and reported both the mean and standard deviation. For the activity cliff estimation task (**TABLE V** and **TABLE VI**), we followed the standard MoleculeACE protocol,

which adopts five-fold cross-validation and reports only mean values, standard deviation was not included in the original paper, and we followed that convention for consistency.

Supplementary Table 1. Details of fine-tuning dataset.

Task	Dataset	Original download link	Description
Reaction Product Prediction	USPTO-479k	MolR	The dataset contains 478,612 chemical reactions, divided into training, validation, and test sets, containing 408,673, 29,973, and 39,966 reactions respectively.
Chemical Reaction Classification	Schneider	DRFP	The dataset contains 38,800 chemical reactions, divided into training, validation, and test sets, containing 31,002, 3,896, and 3,902 reactions respectively.
Molecular Property Prediction	MoleculeNet	KPGT	These datasets, eight are designed for classification tasks and the remaining three are designed for regression tasks. Each cluster was then split into a training set (80%) and a testing set (20%) using stratified random sampling based on the activity cliff label.
Activity Cliff Estimation	MoleculeACE	MoleculeACE	

A.2 Hyperparameters

Following to previous work^{2,3}, we use grid search to find the best combination of hyperparameters on downstream tasks. The specific search spaces are shown in **Supplementary Table 2**.

Supplementary Table 2. Hyperparameters for pretraining and fine-tuning of ReMol.

Hyperparameters	Pretraining	Fine-tuning
Max Epochs	50	[60, 100]
Batch Size	128	[8, 16, 32, 64]
Temperature	0.05	-
Learning Rate	5e-4	[5e-5, 5e-4, 3e-4, 5e-3]
Weight Decay	1e-5	[1e-4, 1e-5]
Momentum	0.9	0.9
Dropout Rate	0.1	[0.1, 0.5]
Learning Rate Decay	Cosine decay	Cosine decay
Image Size	224×224	224×224
Activator	-	[LeakyReLU, GELU]

B. Supplementary Methods

B.1 Time Complexity

The overall runtime of our method can be divided into two main components: data pre-processing and model computation. In summary, the time complexity analysis of the proposed method is shown in **Supplementary Table 3**.

Data Preprocessing: The primary computational cost of data preprocessing involves masking the reactant and product images and calculating the reaction fingerprint.

- ✓ **Masking:** The masking operation depends on the number of atoms and bonds in the reactants and products. Assuming the mask operation itself is $O(I)$, the time complexity of masking is $O(N \times (N_1 + N_2 + N_3 + N_4))$ where N_1 , N_2 , N_3 , and N_4 represent the number of atoms and bonds in the reactants and products, respectively, and N is the number of reactions. Assuming the number of atoms

and bonds in the reactants and products are approximately equal, this simplifies to $O(N \times M)$, where M represents the average number of atoms and bonds.

- ✓ **Reaction Fingerprint:** The time complexity of calculating the reaction fingerprint is related to the number of atoms, which is $O(N \times (N_1 + N_3))$, and similarly simplifies to $O(N \times M)$.

Model Calculation: The model computation complexity is divided into two parts: embedding calculation and pretraining task calculation.

- ✓ **Embedding Calculation:** The model consists of ResNet18 and Transformer components. The time complexity of ResNet18 depends on the input image size and is $O(H \times W \times N \times K^2 \times C^2)$, where H and W are the height and width of the input image, K is the kernel size, and C is the channel. Typically, K and C are constants, so the complexity simplifies to $O(H \times W \times N)$. The time complexity of the Transformer depends on the embedding dimension and is $O(N' \times N' \times D \times N)$, where N' is the sequence length (which can be ignored, as each reactant and product is treated as a token), and D is the embedding dimension. Thus, the time complexity simplifies to $O(N \times D)$.
- ✓ **Pretraining Task:** This paper proposes three pretraining tasks: MRCL, RPCL, and RRCL. The time complexity of MRCL is primarily within the preprocessing and embedding calculation steps, so its complexity is $O(1)$ in the pretraining task calculation. For RPCL, the primary computational cost is the calculation of similarity between all sample pairs in the batch during contrastive learning. The time complexity of this operation is $O(B^2 \times D)$ where B is the batch size. Similarly, RRCL also has a time complexity of $O(B^2 \times D)$.

Supplementary Table 3. The time complexity in the data preprocessing and model calculation, where N is the number of reactions, M represents the average number of atoms and bonds, H and W are the height and width of the input image, D is the embedding dimension and B is the batch size.

Operation	Time Complexity
Data Preprocessing	
Masking	$O(N \times M)$
Reaction Fingerprint	$O(N \times M)$
Model Calculation	
Embedding Calculation	$O(H \times W \times N) + O(N \times D)$
Pretraining Task	$O(B^2 \times D)$

B.2 Computational Requirements

The computational requirements of the proposed method are shown in

Supplementary Table 4.

Supplementary Table 4. The computational requirements in the pretraining and fine-tuning stages. #samples represent the total number of molecules.

(a) The computational requirements in the pretraining stage.				
#samples	#batch	GPU memory	Training time	Server
690K	128	~40G	~1 hours/epoch	CPU: 13th Gen Intel® Core™ i7-13700K GPU: 4090 Ti
(b) The computational requirements in the fine-tuning stage.				
#samples	#batch	GPU memory	Training time	Server
1000	16	2.7G	~0.5 min/epoch	CPU: 13th Gen Intel® Core™ i7-13700K GPU: 4090 Ti

C. Supplementary Results

C.1 Results on Pretraining

In the MRCI task, there are four parts of the loss, namely, mask reactant atom center identification L^R_{atom} , mask reactant bond center identification L^R_{bond} , mask product atom center identification L^P_{atom} , and mask product bond center identification L^P_{bond} . Finally, the loss L of MRCI is as follows: $L = L^R_{atom} + L^R_{bond} + L^P_{atom} + L^P_{bond}$

Finally, the loss of MRCI on the training set is 0.266, and the accuracy on the validation set is shown in **Supplementary Table 5**.

Supplementary Table 5. The accuracy of MRCI task. MRACI represents mask reactant atom center identification task.

Task	MRACI	MRBCI	MPACI	MPBCI
Acc	0.900	0.804	0.953	0.889

C.2 Statistical Significance Analysis

We have conducted additional statistical significance analyses to validate the robustness of our model’s improvements (See **Supplementary Table 6**). ReMol achieves statistically significant differences compared to baselines across all tasks ($p < 0.05$).

Supplementary Table 6. Statistical significance tests of ReMol. We performed significance tests on all four downstream tasks. For molecular property prediction and activity cliff estimation, we randomly selected one dataset.

Dataset	Metrics	Baseline	t-test	U-test	95% Confidence intervals
USPTO-479k	MR ↓	MolR	$<< 0.05$	$<< 0.05$	[-33.53, -11.07]
Schneider	Acc ↑	MolR	0.001	0.001	[0.202,0.238]
Estrogen	AUROC ↑	KPGT	0.046	0.031	[0.001,0.052]
FreeSolv	RMSE ↓	KPGT	0.044	0.045	[-0.032, -0.004]
D4R	RMSE _{cliff} ↓	ImageMol	0.025	0.030	[-0.204, -0.014]

C.3 Dataset Bias Analysis

It is worth clarifying that USPTO-full primarily assesses model generalizability over known reaction types as a benchmark dataset originally designed for single-step retrosynthesis evaluation. Unlike conventional works that directly use it for retrosynthesis prediction, our study innovatively employs it as a pretraining corpus to extract general chemical knowledge (e.g., reaction center) through self-supervised learning. As correctly noted, the absence of explicit reaction type labels (e.g., SN2, Suzuki coupling) in USPTO-full does limit direct quantitative analysis of reaction type distribution.

To rigorously investigate potential bias impacts, we conducted systematic experiments on the Schneider dataset (chemical reaction classification task) by constructing controlled imbalance scenarios. Given its balanced original reaction type distribution, we simulated artificial imbalance by retaining only 10% and 5% of samples from the first five reaction categories while keeping all 41 remaining categories intact. This setup tests whether models disproportionately favor majority reaction types. As shown in **Supplementary Table 7**, both 10% and 5% imbalanced classes caused notable performance degradation (1.3% and 3.2% overall accuracy drops respectively),

particularly in imbalanced categories (9.7% and 25% accuracy decreases). These results confirm that insufficient representation of specific reaction types during training indeed compromises model performance, highlighting the necessity for comprehensive pretraining datasets encompassing diverse reaction types.

Supplementary Table 7. Class U_1 represents the imbalanced class (Sampled Class: 0,1,2,3,4) with each class having only 10% of the full class. Class U_2 represents extremely imbalanced class (Sampled Class: 0,1,2,3,4), with each class having only 5% of the full class. Acc_imbalanced indicates the accuracy of the imbalanced class (Sampled Class: 0,1,2,3,4). The accuracy is computed multiple times on different random samples, and we report the mean and standard deviation results on the test set.

Reaction class in training set	Acc	Acc_imbalanced
Full	0.945(0.001)	0.952(0.001)
5 U_1 classes + 41 Full classes	0.933(0.002)	0.860(0.001)
5 U_2 classes + 41 Full classes	0.915(0.004)	0.714(0.008)

C.4 Few-shot Learning Analysis

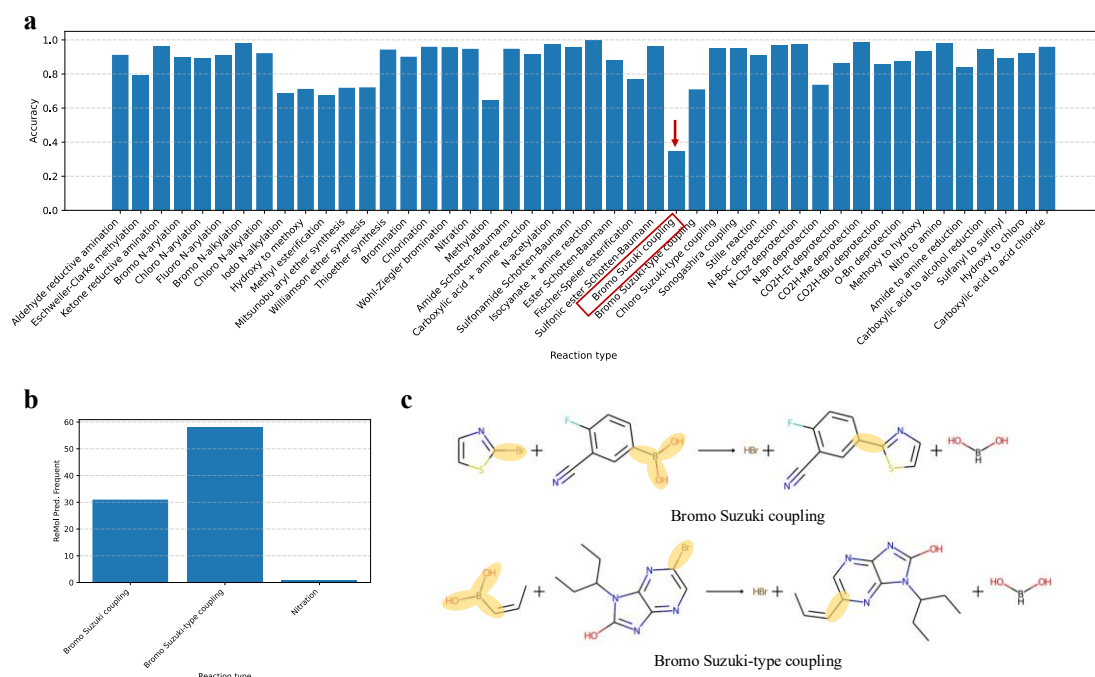
We show the performance of each pretraining task under few-shot conditions in the **Supplementary Table 8**. We attribute this strength of ReMol's small reaction classification dataset capability to two synergistic factors: large-scale pretraining on nearly one million reactions transfers fundamental chemical reaction pattern recognition, while our reaction-aware pretraining tasks, especially Reaction Representation Contrastive Learning (RRCL), distill discriminative reaction-level semantics. RRCL constructs comprehensive reaction fingerprints that integrate both reactant features and product transformations, capturing critical reaction-level features. Crucially, through contrastive learning, RRCL forces the model to cluster chemically similar reactions in the embedding space. This capability enables robust classification even in data-scarce scenarios, as evidenced by RRCL alone boosting small reaction classification dataset accuracy.

Supplementary Table 8. The analysis of ReMol for few-shot reaction classification.

Model	Non-Pretrain	+MRCI	+RPCL	+RRCL	ReMol
32	0.504 _(0.007)	0.697 _(0.003)	0.752 _(0.006)	0.788 _(0.009)	0.816 _(0.010)

C.5 Reaction-type Analysis

To assess whether reaction-center supervision noise varies across reaction families and affects learning unevenly, we performed a reaction-type-specific analysis on the Schneider benchmark. As shown in **Supplementary Figure 2**, the MRCI model achieves consistently high accuracy for many reaction types, suggesting that these classes provide clear and reliable supervisory signals; in contrast, a small subset of classes exhibits markedly lower accuracy, indicating noisier or less consistent reaction-center supervision. A representative example is the Bromo Suzuki coupling, which attains a low accuracy of 0.344 (red arrow in **Supplementary Figure 2a**). Importantly, **Supplementary Figure 2b** reveals that most misclassified samples from this class are predicted as Bromo Suzuki-type coupling, pointing to a systematic confusion rather than random errors. To further verify this, we removed the Bromo Suzuki-type coupling class from evaluation, after which the accuracy on Bromo Suzuki coupling increases sharply to 0.944, confirming that the primary issue is ambiguity between these two closely related classes. As illustrated in **Supplementary Figure 2c**, their reaction centers are largely identical, both follow the Bromo Suzuki-type coupling with C-Br cleavage and C-C bond formation, yet they are assigned to different reaction types because the coupling partners differ: Bromo Suzuki coupling corresponds to an aryl halide reacting with an aryl boronic acid (aryl-aryl coupling), whereas Bromo Suzuki-type coupling covers aryl halides reacting with an alkenyl boronic acid (aryl-alkenyl coupling). This demonstrates that reaction-type labels can diverge even when reaction-center templates are highly similar, effectively introducing label noise at the reaction-type level and degrading apparent per-class accuracy. More broadly, such reaction-dependent ambiguity can weaken MRCI learning for affected families.



Supplementary Figure 2. Reaction-type analysis on the Schneider benchmark. (a) Per-class accuracy of MRCI on Schneider reaction-type prediction, with the red arrow highlighting Bromo Suzuki coupling. (b) Prediction frequency for test samples from the Bromo Suzuki coupling class, showing that many errors are misclassified as Bromo Suzuki-type coupling. (c) Representative examples comparing Bromo Suzuki coupling and Bromo Suzuki-type coupling; the highlighted areas indicate the reaction center.

C.6 Comparison with Multimodal Methods

we have included representative multimodal baselines in our evaluation: GraphMVP [4] and 3DInfomax [5] (both 2D-Graph-3D Geometry) for molecular property prediction and activity cliff estimation, and CGIP [6] (2D Graph-2D Image) as an image-graph multimodal reference in **Supplementary Table 9** and **Supplementary Table 10**. These results demonstrate that ReMol remains competitive even against multimodal methods.

Supplementary Table 9. Comparison of average performance on molecular property prediction with different multi-modal representations.

Modal	Models	Avg AUROC	Avg RMSE
2D Graph-3D Geometry	3DInfomax	0.804	1.400
	GraphMVP	0.769	1.647
2D Graph-2D Image	CGIP-DeeperGCN	0.826	1.251
	CGIP-ResNet18	0.834	1.238
2D Image	ReMol	0.847	1.137

Supplementary Table 10. Comparison of average $\text{RMSE}_{\text{cliff}}$ on activity cliff estimation with different multi-modal representations.

Modal	Models	Avg $\text{RMSE}_{\text{cliff}}$
2D Graph-3D Geometry	3DInfomax	0.903
	GraphMVP	0.905
2D Graph-2D Image	CGIP-DeeperGCN	0.856
	CGIP-ResNet18	0.826
2D Image	ReMol	0.761

C.7 Ablation Study

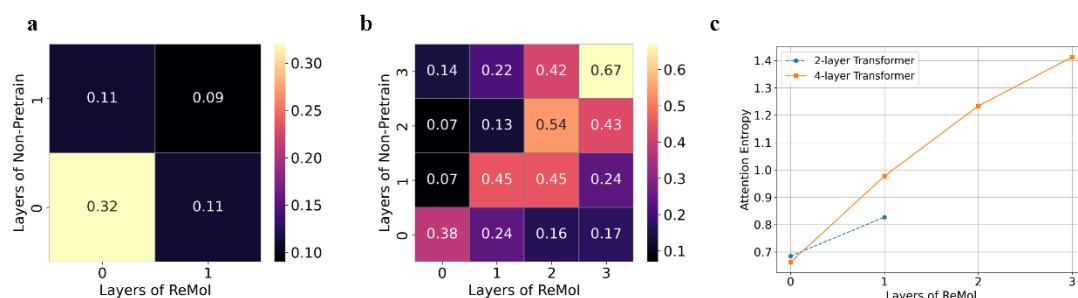
To further validate the effectiveness of our architecture, we conducted additional ablation studies by modifying the backbone network and adjusting the Transformer structure, as shown in **Supplementary Table 11**. Below are the key findings:

- ✓ **Replacing ResNet18 with ResNet50:** We replaced ResNet18 with ResNet50 while keeping the Transformer configuration unchanged (2 layers, 2 heads). Results showed a slight performance improvement, but the training cost increased significantly. Given the marginal gain and high computational overhead, we retained ResNet18 for efficiency and scalability.
- ✓ **Deepening the Transformer (2 $\rightarrow$ 4 layers/heads):** We increased the Transformer depth and heads from 2 to 4 while retaining ResNet18. Surprisingly, this led to performance degradation, likely because the Transformer is primarily employed to address the issue of inconsistent numbers of reactants and products in our task. Unlike natural language processing (NLP), where the number of tokens is much smaller than the sequence length, a deeper Transformer does not necessarily extract the reaction information in our case. This suggests that a

shallower Transformer (2 layers/heads) better balances model capacity and generalization in our task. Furthermore, we conducted additional experiments focusing on attention saturation by analyzing both feature similarity across layers and attention entropy. We computed layer-wise feature similarity between pretrained and non-pretrained models. As shown in **Supplementary Figure 3a**(2-layer model) and **Supplementary Figure 3b** (4-layer model), the deeper model exhibits higher similarity between its top layers and the bottom layers of the non-pretrained model, suggesting a feature collapse or representation degradation phenomenon. We further computed the mean attention entropy at each Transformer layer to quantify attention saturation. **Supplementary Figure 3c** shows that as depth increases, the attention entropy steadily rises, indicating that attention becomes more uniformly distributed, i.e., less focused. This behavior is consistent with attention saturation, where deeper layers tend to lose discriminative focus—potentially leading to degraded performance. These analyses support the notion that simply increasing Transformer depth does not necessarily enhance performance.

Supplementary Table 11. Ablation results of ReMol with different network components on molecular property classification and regression datasets. The bold indicate the results in our original paper.

Model	Classification dataset	Regression dataset
	AVG	AVG
ResNet18+Transformer (with 2 layers and 2 heads)	0.847	1.137
ResNet50+Transformer (with 2 layers and 2 heads)	0.850	1.139
ResNet18+Transformer (with 4 layers and 4 heads)	0.821	1.245

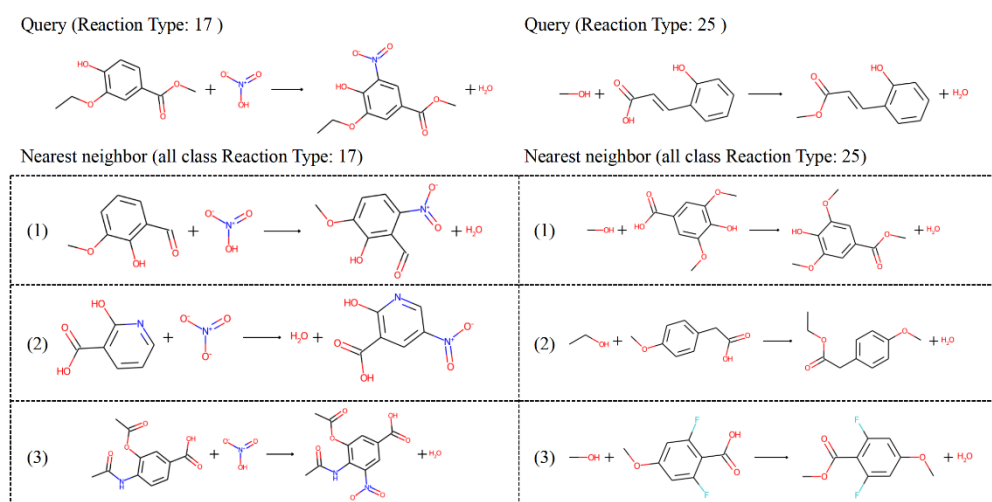


Supplementary Figure 3. Analysis of feature similarity and attention entropy for models with different Transformer depths. (a) Layer-wise feature similarity heatmap between a 2-layer Transformer model and its non-pretrained counterpart. (b) Similarity heatmap for the 4-layer Transformer variant, showing increased similarity in deeper layers, indicating potential representation degradation. (c) Attention entropy curves across Transformer layers.

C.8 Case Study

C.8.1 Similar Reaction Search

Supplementary Figure 4 shows the top3 nearest neighbors found in the test set. All of the retrieved neighbors are of the same class as the query reaction and share the same reaction template. This means that ReMol can utilize template information for reaction classification.



Supplementary Figure 4. A similar reaction search using ReMol features.

C.8.2 Feature Contribution

We conducted additional experiments using Grad-CAM to visualize the attention of the ReMol model. We extend the interpretability analysis by visualizing attention with Grad-CAM across the chemical reaction task and molecular property prediction. For product prediction and reaction classification, ReMol consistently highlights reaction templates and reaction centers (atoms/bonds undergoing changes), as shown in **Supplementary Figure 5**. For molecular property prediction, Grad-CAM concentrates on task-relevant substructures (e.g., hydrophilic, toxic functional groups, activity substructure), as shown in **Supplementary Figure 6**. These observations indicate that the model’s attention aligns with chemically causal regions in reaction tasks and with property-relevant functional groups in property tasks.

- ✓ **Chemical Reaction Explain:** Across both datasets, ReMol allocates the highest saliency to chemically meaningful regions that drive the transformation, rather than to peripheral fragments.

(a) USPTO-479K (reaction product prediction): For each example, the left panel shows the reactant and the right panel the predicted product. Heatmaps concentrate on reaction templates and reaction centers (atoms/bonds undergoing formation, cleavage, or order changes). To quantitatively assess the interpretability of our model and validate whether the attention maps indeed correspond to chemically relevant regions, we conducted a comprehensive evaluation using the USPTO-479k dataset. Specifically, we performed three independent experiments, each time randomly sampling 1,000 reactions from USPTO-479k using different random seeds. The reaction centers were identified through atom mapping, and attention maps for both reactants and products were generated using Grad-CAM. For quantitative evaluation, we proposed two metrics and reported the mean and standard deviation across the three runs to ensure robustness and statistical reliability. **(1) Hit Accuracy:** This binary metric is defined as 1 if the attention map successfully highlights any part of the reaction center, and 0 otherwise. It quantifies whether the model’s attention includes the true reactive atoms (Product: $90.00 \pm 2.21\%$, Reactant: $93.33 \pm 1.13\%$). **(2)**

Attention-Reaction-center Match Ratio (ARMR): which quantifies how

much of the model's focused area corresponds to the actual reactive center, as opposed to unrelated parts of the molecule (Product: $41.54 \pm 3.78\%$, Reactant: $66.61 \pm 2.65\%$). ARMOR is defined as follows:

$$\text{ARMOR} = \frac{|A \cap R|}{|A \cap M|}$$

where A, R, and M represent attention, reaction center, and total molecule area, respectively.

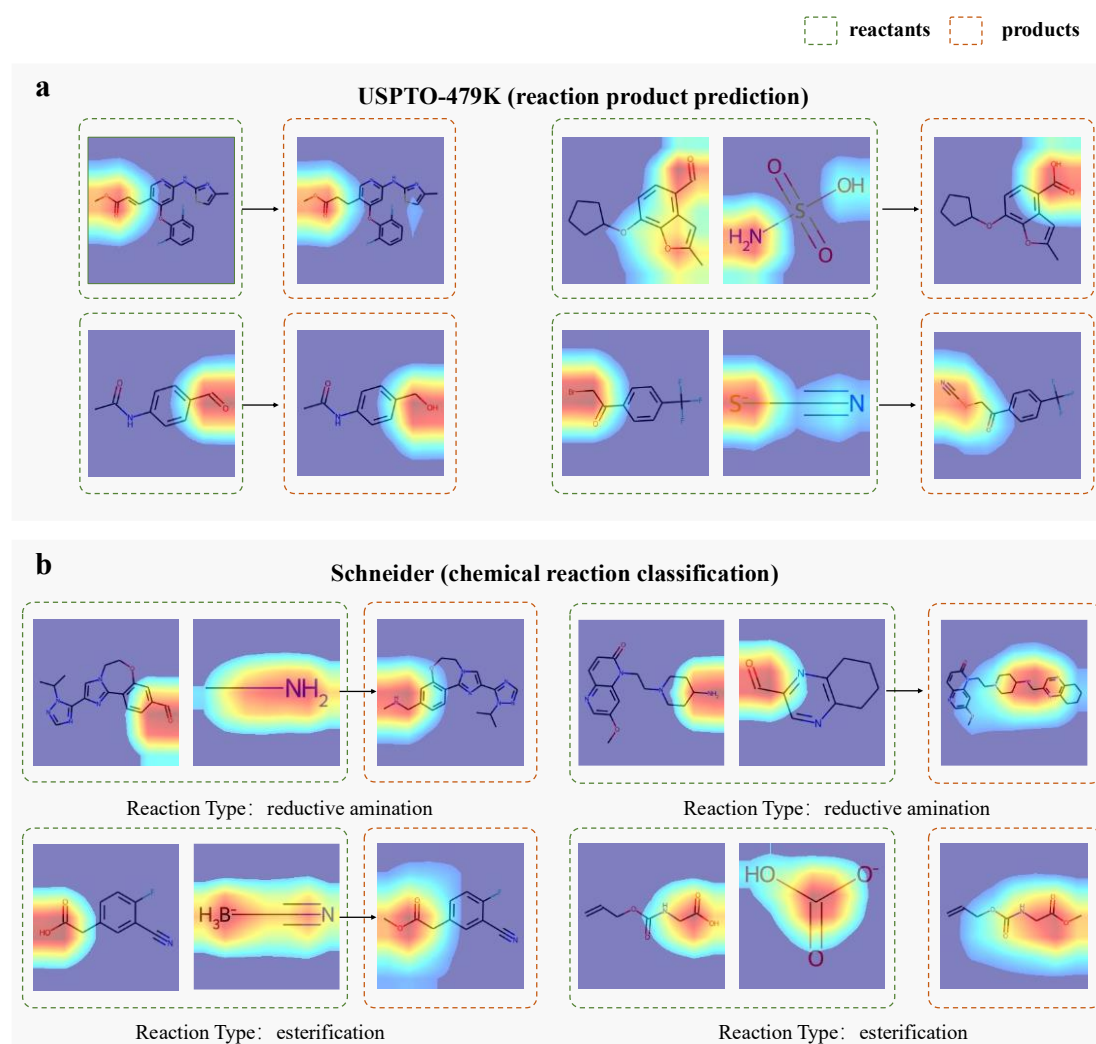
(b) Schneider (reaction classification). For representative classes (e.g., reductive amination, esterification), attention consistently highlights causal substructures (reaction template) that define the class.

✓ **Chemical Property Explain:** Our analysis revealed critical insights into the model's decision-making process, aligning with domain-specific chemical principles:

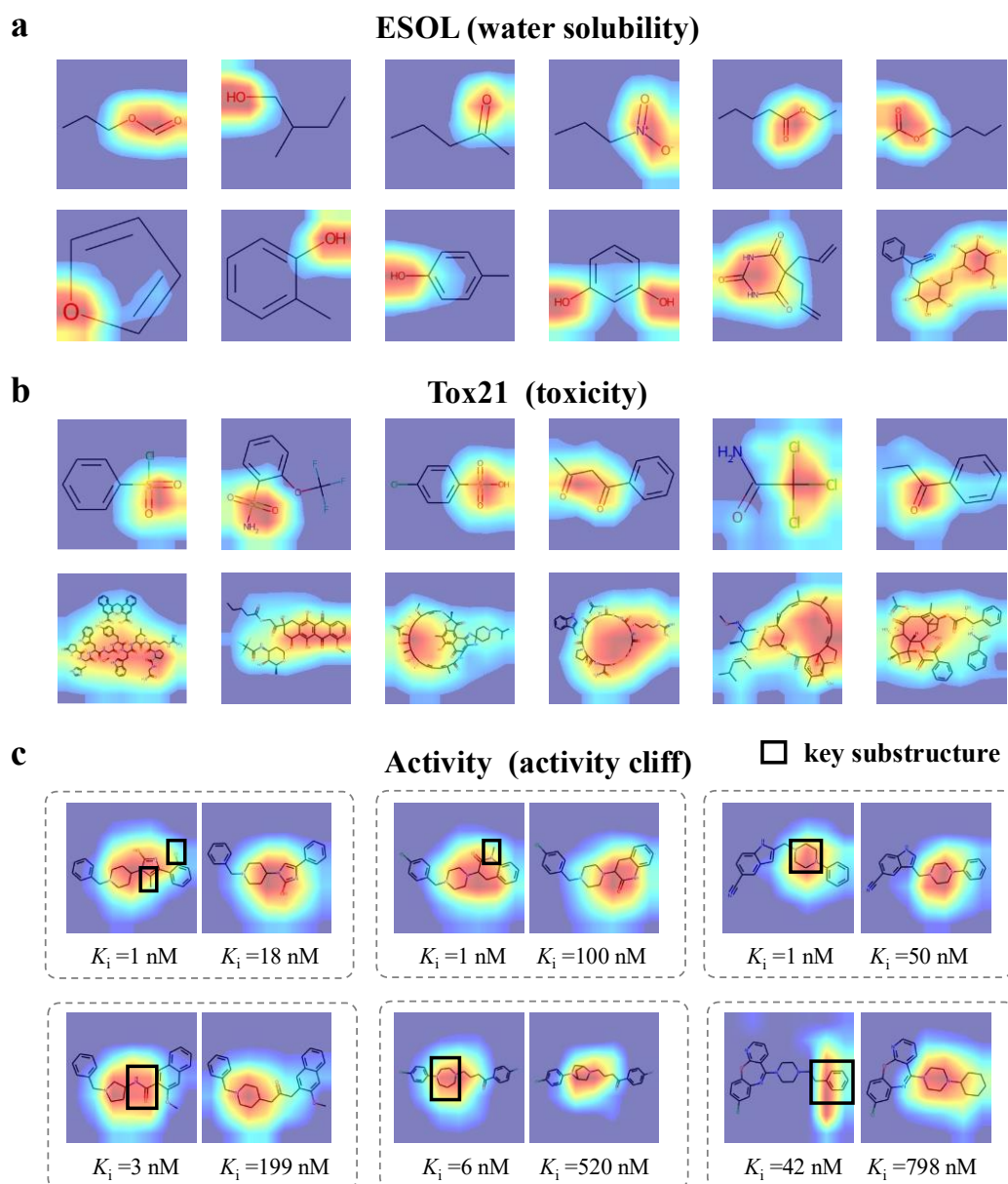
(a) Water Solubility Substructure: The model pays significant attention to polar groups such as hydroxyl (-OH) and carboxyl (-COOH) in the ESOL dataset (**Supplementary Figure 6a**). This is highly consistent with chemical intuition [7], because such groups significantly affect the water solubility of molecules through hydrogen bonding.

(b) Toxicity Substructure: The attention mechanism successfully captures known toxicity-related structures [8] (e.g., benzene rings, macrocyclic compounds) as well as detoxifying groups (e.g., halogen atoms, carboxyl groups) (**Supplementary Figure 6b**).

(c) Activity Cliff Substructure: For molecular cliff pairs, heatmaps pivot to the key substructures that affect changes in biological activity (**Supplementary Figure 6c**).



Supplementary Figure 5. Chemical reaction attention interpretation of ReMol. (a) Examples of ReMol's heatmaps of USPTO-479K dataset (reaction product prediction) by Grad-CAM. (b) Examples of ReMol's heatmaps of Schneider dataset (chemical reaction classification). The green dotted box represents the reactants, and the orange dotted box represents the products.



Supplementary Figure 6. Chemical property attention interpretation of ReMol. (a) Examples of ReMol's heatmaps of ESOL dataset by Grad-CAM. (b) Examples of ReMol's heatmaps of Tox21 dataset. (c) Examples of ReMol's heatmaps of D4R dataset (activity cliff estimation). The black borders indicate the key substructures that affect changes in biological activity.

Supplementary Table 12. Comparison of RMSE on activity cliff estimation based on different methods.

Method	D4R	DAT	FX	GSK3	HRH3	SOR	ABL11	GR	CLK4	OX2R	AVG
LSTM	0.819	0.675	0.783	0.743	0.651	0.818	0.732	0.806	0.942	0.729	0.770
ChemBERTa	0.898	0.814	1.027	0.874	0.755	0.939	0.818	0.734	0.968	0.872	0.870
CNN	1.006	0.913	1.161	0.940	0.897	0.990	0.996	0.830	1.009	0.941	0.968
AFP	0.974	0.935	1.016	0.897	0.735	0.863	0.835	0.842	1.014	0.870	0.898
MPNN	1.047	0.816	1.208	0.913	0.849	0.958	0.813	0.846	1.028	0.919	0.940
GAT	1.085	0.988	1.458	0.936	0.894	1.094	0.877	0.838	1.041	0.926	1.014
GCN	1.108	0.935	1.258	0.952	0.807	0.964	0.867	0.852	0.974	0.990	0.971
Edgepred	0.924	0.747	0.958	0.952	0.754	0.844	0.965	0.903	0.959	0.858	0.886
Mole-BERT	0.882	0.719	0.909	0.815	0.770	0.878	0.797	0.845	0.852	0.822	0.829
3DInfomax	0.793	0.932	1.004	1.048	0.714	0.775	0.753	0.776	0.970	0.805	0.857
GraphMVP	0.844	0.713	0.882	0.837	0.716	0.827	0.782	0.830	0.966	0.781	0.818
MolR	0.896	0.768	0.976	0.821	0.711	0.863	0.780	0.833	0.878	0.800	0.833
ImageMol	0.858	0.694	0.958	0.817	0.686	0.806	0.770	0.804	0.928	0.812	0.813
ReMol	0.795	0.616	0.769	0.719	0.610	0.779	0.695	0.732	0.887	0.710	0.731

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